VAYUSETU: System Architecture Document

Production Tech Stack, Database Schema, & API Contracts

1. Production Technology Stack

- **Frontend Command Center:** Next.js 14, React 18, TypeScript, Tailwind CSS, and Leaflet JS for GIS map rendering.
- **Backend Gateway:** FastAPI (Python 3.11) with structured loggers and asynchronous routing, running under Uvicorn.
- **AI Framework:** PyTorch 2.2 with CPU/GPU acceleration, scikit-learn for regression ensembles, and pandas/numpy for data handling.
- **Database Schema:** SQLite database for fast local execution (pilot), migrating to high-performance PostgreSQL (production) with PostGIS extensions for geospatial query optimization.

2. Production Database Schema

The SQLite database (`vayusetu.db`) contains key relational tables structured as follows:

- **climate_state:** Columns: id (PK), timestamp, district (INDEX), temperature, rainfall, soil_moisture, humidity, lst, sst, albedo, and calculated CRI score. Stores versioned state data.
- **lineage_records:** Columns: id (PK), version, timestamp, run_id, sources, coverage_pct, confidence_score, and operator. Tracks dataset heritage.
- **audit_trail:** Columns: id (PK), timestamp, event_type, details, and user_id. Immutable logging table.

3. REST API Contract Matrix

Endpoint	Method	Query Params	Response Keys	Status
/api/v1/climate/live-state	GET	district	temperature, rainfall, soil_moisture, sst, lst, CRI	Active
/api/v1/prediction/forecast	GET	district	ensemble_prediction, confidence_pct, range_bounds	Active
/api/v1/simulation/runoff	GET	precipitation, urbanization, forest	scenario_studio_metrics, district_breakdown	Active
/api/v1/twin/twin-lineage	GET	version	twin_version, run_id, dataset_sources, coverage_pct	New API
/api/v1/twin/time-travel	GET	offset_hours	timestamp, simulated_state, risk_scores	New API
/api/v1/twin/decision-impact	GET	district	affected_districts, affected_population, actions	New API